AGCO – Prédio ACADEMY

Avenida José Alves de Oliveira, Nº 300,
Jundiaí-SP

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REVISÃO | 00
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ARQUITETURA • ENGENHARIA

MEMÓRIAL DE CÁLCULO DE HVAC

MEMÓRIAL DE CÁLCULO

ENGENHARIA DE AR-CONDICIONADO

AGCO – Prédio Academy

Avenida José Alves de Oliveira, Nº300, Jundiaí-SP

SUMÁRIO

1. OBJETIVO....4

2. NORMAS APLICÁVEIS....4

3. DADOS DE ENTRADA....5

3.1 LOCALIZAÇÃO....5

3.2 CONDIÇÕES PSICROMÉTRICAS....5

3.2.1 Dados da norma ABNT NBR 16401-3:2008 5

3.2.2 Condições do ar interno refrigerado....6

3.3 COEFICIENTES DE SEGURANÇA ADOTADOS....6

3.4 FATOR DE BY-PASS DAS SERPENTINAS....6

3.5 NÍVEL DE RUÍDO....6

3.6 COEFICIENTES DE TRANSMISSÃO....6

3.7 DISSIPAÇÃO TÉRMICA DE ILUMINAÇÃO E EQUIPAMENTOS....7

3.8 DISSIPAÇÃO TÉRMICA DEVIDO A PRESENÇA HUMANA....7

4. ZONEAMENTO DAS ZONAS TÉRMICAS....8

5. VAZÃO DE AR EXTERIOR....9

6. VAZÃO DE EXAUSTÃO....11

7. CARGA TÉRMICA....12

7.1 RESUMO DOS VALORES DE CARGAS TÉRMICAS....12

Projeto: AGCO – Prédio Academy

1. OBJETIVO

O presente documento tem por objetivo apresentar a memória de cálculo das cargas térmicas do sistema de ar-condicionado do prédio Academy da AGCO Corporation, localizado no Galpão 18, na Avenida José Alves de Oliveira, Nº 300, Jundiaí-SP.

Para o cálculo dos requisitos de ar exterior e exaustão, foram utilizadas planilhas do aplicativo Microsoft Excel. Já para o cálculo da carga térmica, foram empregadas planilhas do Excel em conjunto com o programa '*Hourly Analysis Program (HAP)*', versão 4.90 da Carrier Corporation.

2. NORMAS APLICÁVEIS

Todos os cálculos envolvidos, dimensionamento, especificações, bem como fornecimento e instalação, devem obedecer às normas, regulamentos e publicações técnicas a seguir:

- NBR 15220 – Desempenho térmico de edificações;
- NBR-16401-1 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 1 – Projetos das instalações;
- NBR-16401-2 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 2 – Parâmetros de conforto térmico;
- NBR-16401-3 – Instalações de ar-condicionado – Sistemas centrais e unitários – Parte 3 – Qualidade do ar interior.
- NBR-10151 – Acústica-Avaliação do Ruído em áreas habitadas.
- NBR-5410 – Instalações elétricas baixas tensão.
- NR (Normas Regulamentares) – Ministério do Trabalho;
- Regulamento GM/MS nº3523/1998 – Ministério da Saúde;
- Resolução RE-09: 2003 da ANVISA;
- Decreto n.º 22281 de 19 de novembro 2002 - Regulamento para a Instalação e Conservação de Sistemas de Ar-Condicionado e Ventilação Mecânica no Município do Rio de Janeiro.
- AMCA – "Air Movement and Control Association".
- Publicações técnicas da ASHRAE – "American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc".
- SMACNA – "Sheet Metal and Air Conditioning Contractors National Association".

3. DADOS DE ENTRADA

Este capítulo apresenta uma descrição detalhada dos parâmetros de entrada utilizados no projeto. Isso inclui informações climáticas específicas do local do projeto, dados psicométricos relevantes, fator de segurança adotado e outros parâmetros essenciais utilizados no cálculo da carga térmica.

3.1 LOCALIZAÇÃO

Cidade: São Paulo

Referência: Campinas

Altitude: 661 m

Latitude: 23,00º Sul

Longitude: 47,13º Oeste

3.2 CONDIÇÕES PSICROMÉTRICAS

3.2.1 Dados da norma ABNT NBR 16401-3:2008

Segundo a norma ABNT NBR 16401-3:2008, as condições psicrométricas do ar externo para o percentil 0,4 dos dias mais quentes do ano e para o percentil 99,6 dos dias mais frios do ano são:

3.2.1.1 RESFRIAMENTO:

Referência: Campinas - SP

Mês de maior temperatura: fevereiro

TBS Máximo: 33,2 ºC

TBU Coincidente: 21,9 ºC

Média mensal da variação de temperatura diária (ΔT): 9,8 ºC

3.2.2 Condições do ar interno refrigerado

A carga térmica será calculada com o objetivo de manter o ar interno refrigerado no intervalo de temperatura a seguir:

Ar Interno (TBS/UR)[1]: 22,0 °C ± 1,0 °C

3.3 COEFICIENTES DE SEGURANÇA ADOTADOS

Calor sensível: 10%

Calor latente: 10%

3.4 FATOR DE BY-PASS DAS SERPENTINAS

Foi considerado fator de by-pass das serpentinas de 15%

3.5 NÍVEL DE RUÍDO

Os níveis de ruído permitidos em cada área foram determinados com base nas recomendações das normas ABNT NBR-16401-1, item 7.4 e ABNT NBR-10152, tabela I. Além disso, essas normas estão em conformidade com as normas técnicas da ASHRAE e SMACNA. As normas que seguimos são listadas a seguir:

- ABNT NBR 16401
- ABNT NBR 10152
- ASHRAE Handbook – Application – 2019 – Capítulo 49
- ASHRAE Handbook – Fundamentals – 2017 – Capítulo 8
- SMACNA – 1990 – Chapter 11 – HVAC Systems duct design.

3.6 COEFICIENTES DE TRANSMISSÃO

Parede interna: 2,54 W/m² ºC

Parede externa: 2,31 W/m² ºC (coeficiente de sombreamento de 90%)

Laje interna: 2,20 W/m² ºC

Vidro Simples (Comuns): 5,9 W/m² ºC (coeficiente de sombreamento de 0,7)

[1] Temperatura de conforto conforme recomendado na seção 5 da norma ABNT NBR 16401-2:2008.

3.7 DISSIPAÇÃO TÉRMICA DE ILUMINAÇÃO E EQUIPAMENTOS

Os valores adotados para a carga térmica de iluminação e equipamentos foram baseados nas recomendações da NBR 16401-1, em conjunto com a análise da planta baixa de arquitetura fornecida. Para os equipamentos que não são especificados nas normas, recorremos a valores padrão de equipamentos semelhantes quando a especificação está disponível, ou com base em similares disponíveis no mercado nacional.

3.8 DISSIPAÇÃO TÉRMICA DEVIDO A PRESENÇA HUMANA

Sentado, trabalho leve:

Carga sensível: 70 W/pessoa (NBR 16401-1)

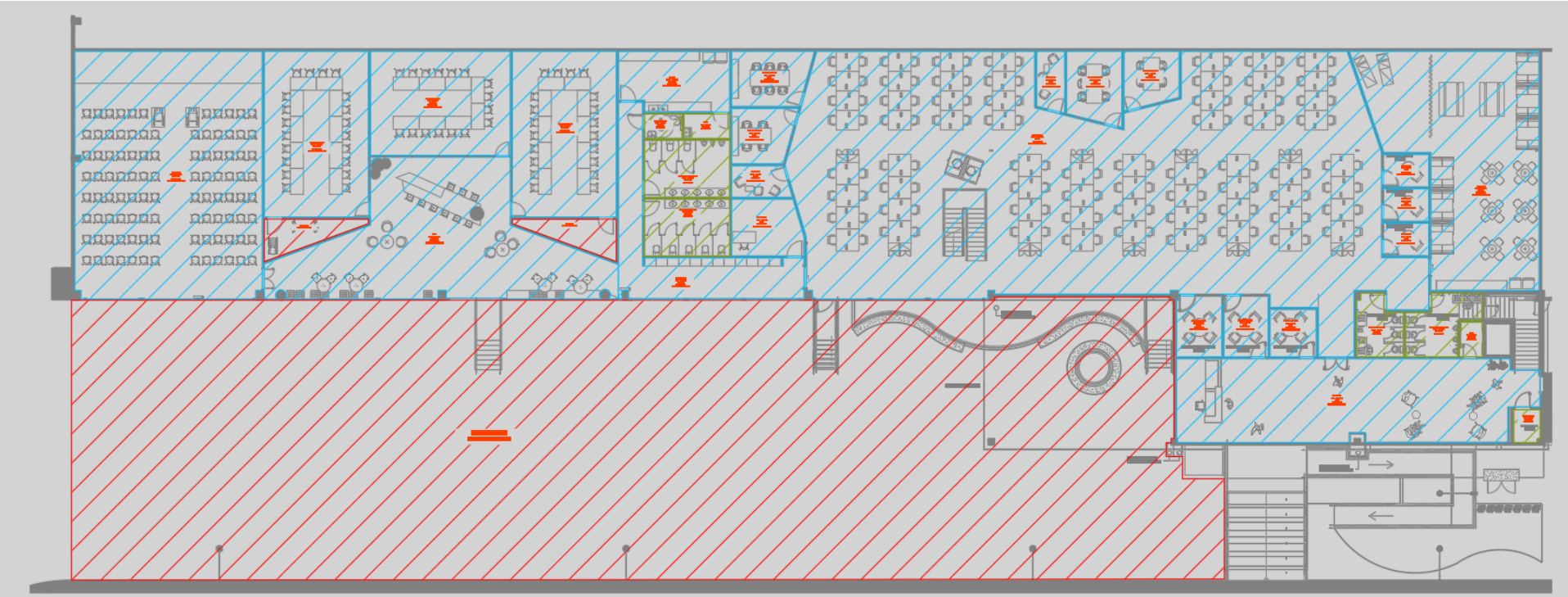
Carga latente: 45 W/pessoa (NBR 16401-1)

Parado em pé, trabalho moderado; caminhando:

Carga sensível: 75 W/pessoa (NBR 16401-1)

Carga latente: 55 W/pessoa (NBR 16401-1)

4. ZONEAMENTO DAS ZONAS TÉRMICAS



LEGENDA DE ZONAS TÉRMICAS - HVAC
ZONA TÉRMICA DE INSUFLAMENTO DE AR CLIMATIZADO (ANSI 31 HATCH)
ZONA TÉRMICA DE EXAUSTÃO (ANSI 31 HATCH)
ZONA TÉRMICA DE FLUXO DE AR EXTERIOR (ANSI 31 HATCH)
ZONA SEM CLIMATIZAÇÃO E/OU NÃO INTERVENÇÃO (ANSI 31 HATCH)

5. VAZÃO DE AR EXTERIOR

As taxas de ar exterior foram calculadas conforme requerimentos da ANVISA e ABNT NBR 16401-3:2008 (NÍVEL 2)

Ambientes	Área	Pessoas	Requisito por pessoa ANVISA	Requisito por pessoa ABNT	Requisito por área ABNT	Vazão de AE MÍNIMA
Auditório 1	175.06 m^2	117 Pessoas	7.50 L/s por pessoa	6.30 L/s por pessoa	0.40 L/s por m^2	3159.0 m^3/h
Sala de reunião 4	65.34 m^2	20 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	540.0 m^3/h
Sala de reunião 5	55.13 m^2	16 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	432.0 m^3/h
Hall 1	118.01 m^2	23 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	621.0 m^3/h
Sala de reunião 6	65.34 m^2	20 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	540.0 m^3/h
Circulação 1	42.51 m^2	7 Pessoas	7.50 L/s por pessoa	5.30 L/s por pessoa	0.40 L/s por m^2	194.8 m^3/h
Sala de reunião 7	15.75 m^2	6 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	162.0 m^3/h
Sala de reunião 8	12.41 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Descanso 1	6.69 m^2	3 Pessoas	7.50 L/s por pessoa	4.80 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Sala geral 1	14.76 m^2	6 Pessoas	7.50 L/s por pessoa	6.30 L/s por pessoa	0.80 L/s por m^2	178.6 m^3/h
Workplace 1	514.39 m^2	130 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	3510.0 m^3/h
Descanso 2	6.19 m^2	3 Pessoas	7.50 L/s por pessoa	4.80 L/s por pessoa	0.40 L/s por m^2	81.0 m^3/h
Sala de reunião 9	14.51 m^2	6 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	162.0 m^3/h

Sala de reunião 10	11.11 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Call room 1	5.58 m^2	1 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	27.0 m^3/h
Call room 2	5.58 m^2	1 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	27.0 m^3/h
Call room 3	5.58 m^2	1 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	27.0 m^3/h
Refeitório 1	122.64 m^2	60 Pessoas	7.50 L/s por pessoa	4.80 L/s por pessoa	1.10 L/s por m^2	1620.0 m^3/h
Sala de reunião 1	10.42 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Sala de reunião 2	10.17 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Sala de reunião 3	8.32 m^2	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Recepção 1	109.01 m^2	11 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	297.0 m^3/h
Copa	24.45	4 Pessoas	7.50 L/s por pessoa	3.10 L/s por pessoa	0.40 L/s por m^2	108.0 m^3/h
Total						**12307.4 m^3/h**

6. VAZÃO DE EXAUSTÃO

As taxas de vazão de exaustão foram calculadas conforme requerimentos da ANVISA e ABNT NBR 16401-3:2008

Banheiros	Área	Pé Direito	# bacias ou chuveiros	Renovações por hora ANVISA	Requisito por bacia ABNT	Vazão de exaustão mínima
Banheiro PNE 1	3.48 m^2	2,70 m	1 unidades	15 renovações	35 L/s por bacia	152 m^3/h
DML 1	4.24 m^2	2,70 m	1 unidades	15 renovações	-	172 m^3/h
Banheiro masculino 1	18.34 m^2	2,70 m	5 unidades	15 renovações	35 L/s por bacia	750 m^3/h
Banheiro feminino 1	18.33 m^2	2,70 m	5 unidades	15 renovações	35 L/s por bacia	750 m^3/h
Copa	23,0 m^2	2,70 m	-	10 renovações	-	621 m^3/h
Refeitório	110,0 m^2	3,5 m	-	8 renovações	-	3080 m^3/h

7. CARGA TÉRMICA

7.1 RESUMO DOS VALORES DE CARGAS TÉRMICAS

Ambiente	Vazão de ar insuflado L/s	Vazão de ar insuflado m^3/h	Carga térmica em kW	Carga térmica em TR	Carga térmica em BTU/h
Auditório 1	2120 L/s	7632 m^3/h	43.60 kW	12.40 TR	148 769 BTU/h
Call room 1	20 L/s	72 m^3/h	0.40 kW	0.11 TR	1 365 BTU/h
Call room 2	20 L/s	72 m^3/h	0.40 kW	0.11 TR	1 365 BTU/h
Call room 3	20 L/s	72 m^3/h	0.40 kW	0.11 TR	1 365 BTU/h
Circulação 1	1396 L/s	5026 m^3/h	18.90 kW	5.37 TR	64 489 BTU/h
Copa 1	509 L/s	1832 m^3/h	7.10 kW	2.02 TR	24 226 BTU/h
Descanso 1	53 L/s	191 m^3/h	1.20 kW	0.34 TR	4 095 BTU/h
Descanso 2	58 L/s	209 m^3/h	1.30 kW	0.37 TR	4 436 BTU/h
Hall 1	1721 L/s	6196 m^3/h	22.20 kW	6.31 TR	75 750 BTU/h
Recepção 1	1513 L/s	5447 m^3/h	18.90 kW	5.37 TR	64 489 BTU/h
Refeitório 1	1063 L/s	3827 m^3/h	23.40 kW	6.65 TR	79 844 BTU/h
Sala de Reunião 1	206 L/s	742 m^3/h	2.90 kW	0.82 TR	9 895 BTU/h
Sala de Reunião 10	146 L/s	526 m^3/h	2.40 kW	0.68 TR	8 189 BTU/h
Sala de Reunião 2	77 L/s	277 m^3/h	1.70 kW	0.48 TR	5 801 BTU/h
Sala de Reunião 3	76 L/s	274 m^3/h	1.70 kW	0.48 TR	5 801 BTU/h
Sala de Reunião 4	627 L/s	2257 m^3/h	11.40 kW	3.24 TR	38 898 BTU/h
Sala de Reunião 5	394 L/s	1418 m^3/h	7.90 kW	2.25 TR	26 956 BTU/h
Sala de Reunião 6	580 L/s	2088 m^3/h	10.80 kW	3.07 TR	36 851 BTU/h
Sala de Reunião 7	214 L/s	770 m^3/h	3.60 kW	1.02 TR	12 284 BTU/h
Sala de Reunião 8	113 L/s	407 m^3/h	2.10 kW	0.60 TR	7 165 BTU/h
Sala de Reunião 9	190 L/s	684 m^3/h	3.40 kW	0.97 TR	11 601 BTU/h
Sala geral 1	121 L/s	436 m^3/h	2.60 kW	0.74 TR	8 872 BTU/h
Workplace 1	4445 L/s	16002 m^3/h	74.80 kW	21.27 TR	255 228 BTU/h

Air System Sizing Summary for AC-01

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

1. AIR SYSTEM INFORMATION

Air System Name	**AC-01**	
Equipment Class	**SPLT AHU**	
Air System Type	**SZCAV**	
Number of zones	**1**	
Floor Area	**175,1**	m²
Location	**Sao Paulo, Brazil**	

2. Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

3. Central Cooling Coil Sizing Data

Total coil load	**43,6**	kW	Load occurs at	**Sep 1500**	
Sensible coil load	**31,9**	kW	OA DB / WB	**32,1 / 21,3**	°C
Coil L/s at Sep 1500	**2120**	L/s	Entering DB / WB	**26,6 / 18,5**	°C
Max block L/s	**2120**	L/s	Leaving DB / WB	**13,1 / 12,1**	°C
Sum of peak zone L/s	**2120**	L/s	Coil ADP	**10,8**	°C
Sensible heat ratio	**0,732**		Bypass Factor	**0,150**	
m²/kW	**4,0**		Resulting RH	**53**	%
W/m²	**248,8**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

4. Supply Fan Sizing Data

Actual max L/s	**2120**	L/s	Fan motor BHP	**1,48**	BHP
Standard L/s	**1959**	L/s	Fan motor kW	**1,18**	kW
Actual max L/(s-m²)	**12,11**	L/(s-m²)	Fan static	**300**	Pa

5. Outdoor Ventilation Air Data

Design airflow L/s	**878**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**5,01**	L/(s-m²)			

Zone Sizing Summary for AC-01

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

6. Air System Information

Air System Name **AC-01**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **175,1** m²
Location **Sao Paulo, Brazil**

7. Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

8. Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	20,9	2077	2077	Oct 1500	5,8	175,1	11,86

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Auditório 1	1	20,9	Oct 1500	2077	5,8	175,1	11,86

Air System Design Load Summary for AC-01

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Sep 1500 COOLING OA DB / WB 32,1 °C / 21,3 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	16 m²	1934	-	16 m²	-	-
Wall Transmission	22 m²	736	-	22 m²	581	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	16 m²	735	-	16 m²	980	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	92 m²	2544	-	92 m²	1399	-
Ceiling	175 m²	4202	-	175 m²	2311	-
Overhead Lighting	2101 W	1142	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1702 W	1532	-	0	0	-
People	117	5687	4118	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	1851	412	10%	527	0
>> Total Zone Loads	**-**	**20364**	**4530**	**-**	**5799**	**0**
Zone Conditioning	-	20296	4530	-	-1038	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	630	-	0	0	-
Return Fan Load	2120 L/s	0	-	2120 L/s	0	-
Ventilation Load	878 L/s	9156	7159	878 L/s	2107	0
Supply Fan Load	2120 L/s	1178	-	2120 L/s	-1178	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	611	-	3%	174	-
>> Total System Loads	**-**	**31871**	**11689**	**-**	**65**	**0**
Central Cooling Coil	-	31871	11690	-	0	0
>> Total Conditioning	**-**	**31871**	**11690**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

September DESIGN COOLING DAY, 1500

9. TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,1	0,01283	878	400	9156	7159
Vent - Return Mixing	Outlet	26,6	0,01107	2120	755	-	-
Central Cooling Coil	Outlet	13,1	0,00905	2120	755	31871	11690
Supply Fan	Outlet	13,6	0,00905	2120	755	1178	-
Cold Supply Duct	Outlet	13,9	0,00905	2077	755	-	-
Zone Air	-	22,6	0,00985	2077	1011	20296	4530
Return Plenum	Outlet	22,9	0,00985	2077	1011	630	-
Duct Leakage Air	Outlet	13,6	0,00905	42	755	-	-
Return Duct	Outlet	22,7	0,00983	2120	1006	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s)
Site Altitude = 661,0 m

10. TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	20364	Cooling	20296	22,6	2077	1011	0	0

Air System Sizing Summary for AC-02

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-02**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**5,6**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0,4**	kW	Load occurs at	**Jul 1400**	
Sensible coil load	**0,3**	kW	OA DB / WB	**32,9 / 21,8**	°C
Coil L/s at Jul 1400	**20**	L/s	Entering DB / WB	**27,0 / 18,9**	°C
Max block L/s	**20**	L/s	Leaving DB / WB	**13,3 / 12,2**	°C
Sum of peak zone L/s	**20**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,713**		Bypass Factor	**0,150**	
m²/kW	**12,9**		Resulting RH	**55**	%
W/m²	**77,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**20**	L/s	Fan motor BHP	**0,01**	BHP
Standard L/s	**19**	L/s	Fan motor kW	**0,01**	kW
Actual max L/(s-m²)	**3,61**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**8**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**1,34**	L/(s-m²)			

Zone Sizing Summary for AC-02

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-02**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **5,6** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,2	20	20	Jan 1500	0,0	5,6	3,54

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Call room 1	1	0,2	Jan 1500	20	0,0	5,6	3,54

Air System Design Load Summary for AC-02

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1400 COOLING OA DB / WB 32,9 °C / 21,8 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	67 W	36	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	103 W	92	-	0	0	-
People	1	50	60	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	18	6	10%	0	0
>> Total Zone Loads	**-**	**196**	**66**	**-**	**0**	**0**
Zone Conditioning	-	194	66	-	-20	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	20	-	0	0	-
Return Fan Load	20 L/s	0	-	20 L/s	0	-
Ventilation Load	8 L/s	79	58	8 L/s	32	0
Supply Fan Load	20 L/s	11	-	20 L/s	-11	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	6	-	3%	0	-
>> Total System Loads	**-**	**310**	**125**	**-**	**1**	**0**
Central Cooling Coil	-	310	125	-	0	0
>> Total Conditioning	**-**	**310**	**125**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-02

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,9	0,01321	8	400	79	58
Vent - Return Mixing	Outlet	27,0	0,01141	20	863	-	-
Central Cooling Coil	Outlet	13,3	0,00915	20	863	310	125
Supply Fan	Outlet	13,8	0,00915	20	863	11	-
Cold Supply Duct	Outlet	14,0	0,00915	20	863	-	-
Zone Air	-	22,8	0,01037	20	1142	194	66
Return Plenum	Outlet	23,7	0,01037	20	1142	20	-
Duct Leakage Air	Outlet	13,8	0,00915	0	863	-	-
Return Duct	Outlet	23,5	0,01035	20	1136	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	196	Cooling	194	22,8	20	1142	0	0

Air System Sizing Summary for AC-03

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-03**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**5,6**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0,4**	kW	Load occurs at	**Jul 1400**	
Sensible coil load	**0,3**	kW	OA DB / WB	**32,9 / 21,8**	°C
Coil L/s at Jul 1400	**20**	L/s	Entering DB / WB	**27,0 / 18,9**	°C
Max block L/s	**20**	L/s	Leaving DB / WB	**13,3 / 12,2**	°C
Sum of peak zone L/s	**20**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,713**		Bypass Factor	**0,150**	
m²/kW	**12,9**		Resulting RH	**55**	%
W/m²	**77,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**20**	L/s	Fan motor BHP	**0,01**	BHP
Standard L/s	**19**	L/s	Fan motor kW	**0,01**	kW
Actual max L/(s-m²)	**3,61**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**8**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**1,34**	L/(s-m²)			

Zone Sizing Summary for AC-03

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-03**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **5,6** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,2	20	20	Jan 1500	0,0	5,6	3,54

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Call room 2	1	0,2	Jan 1500	20	0,0	5,6	3,54

Air System Design Load Summary for AC-03

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1400 COOLING OA DB / WB 32,9 °C / 21,8 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	67 W	36	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	103 W	92	-	0	0	-
People	1	50	60	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	18	6	10%	0	0
>> Total Zone Loads	**-**	**196**	**66**	**-**	**0**	**0**
Zone Conditioning	-	194	66	-	-20	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	20	-	0	0	-
Return Fan Load	20 L/s	0	-	20 L/s	0	-
Ventilation Load	8 L/s	79	58	8 L/s	32	0
Supply Fan Load	20 L/s	11	-	20 L/s	-11	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	6	-	3%	0	-
>> Total System Loads	**-**	**310**	**125**	**-**	**1**	**0**
Central Cooling Coil	-	310	125	-	0	0
>> Total Conditioning	**-**	**310**	**125**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

July DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,9	0,01321	8	400	79	58
Vent - Return Mixing	Outlet	27,0	0,01141	20	863	-	-
Central Cooling Coil	Outlet	13,3	0,00915	20	863	310	125
Supply Fan	Outlet	13,8	0,00915	20	863	11	-
Cold Supply Duct	Outlet	14,0	0,00915	20	863	-	-
Zone Air	-	22,8	0,01037	20	1142	194	66
Return Plenum	Outlet	23,7	0,01037	20	1142	20	-
Duct Leakage Air	Outlet	13,8	0,00915	0	863	-	-
Return Duct	Outlet	23,5	0,01035	20	1136	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	196	Cooling	194	22,8	20	1142	0	0

Air System Sizing Summary for AC-04

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-04**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**5,6**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0,4**	kW	Load occurs at	**Jul 1400**	
Sensible coil load	**0,3**	kW	OA DB / WB	**32,9 / 21,8**	°C
Coil L/s at Jul 1400	**20**	L/s	Entering DB / WB	**27,0 / 18,9**	°C
Max block L/s	**20**	L/s	Leaving DB / WB	**13,3 / 12,2**	°C
Sum of peak zone L/s	**20**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,713**		Bypass Factor	**0,150**	
m²/kW	**12,9**		Resulting RH	**55**	%
W/m²	**77,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**20**	L/s	Fan motor BHP	**0,01**	BHP
Standard L/s	**19**	L/s	Fan motor kW	**0,01**	kW
Actual max L/(s-m²)	**3,61**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**8**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**1,34**	L/(s-m²)			

Zone Sizing Summary for AC-04

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-04**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **5,6** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,2	20	20	Jan 1500	0,0	5,6	3,54

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Call room 3	1	0,2	Jan 1500	20	0,0	5,6	3,54

Air System Design Load Summary for AC-04

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1400 COOLING OA DB / WB 32,9 °C / 21,8 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	67 W	36	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	103 W	92	-	0	0	-
People	1	50	60	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	18	6	10%	0	0
>> Total Zone Loads	**-**	**196**	**66**	**-**	**0**	**0**
Zone Conditioning	-	194	66	-	-20	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	20	-	0	0	-
Return Fan Load	20 L/s	0	-	20 L/s	0	-
Ventilation Load	8 L/s	79	58	8 L/s	32	0
Supply Fan Load	20 L/s	11	-	20 L/s	-11	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	6	-	3%	0	-
>> Total System Loads	**-**	**310**	**125**	**-**	**1**	**0**
Central Cooling Coil	-	310	125	-	0	0
>> Total Conditioning	**-**	**310**	**125**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

July DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,9	0,01321	8	400	79	58
Vent - Return Mixing	Outlet	27,0	0,01141	20	863	-	-
Central Cooling Coil	Outlet	13,3	0,00915	20	863	310	125
Supply Fan	Outlet	13,8	0,00915	20	863	11	-
Cold Supply Duct	Outlet	14,0	0,00915	20	863	-	-
Zone Air	-	22,8	0,01037	20	1142	194	66
Return Plenum	Outlet	23,7	0,01037	20	1142	20	-
Duct Leakage Air	Outlet	13,8	0,00915	0	863	-	-
Return Duct	Outlet	23,5	0,01035	20	1136	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	196	Cooling	194	22,8	20	1142	0	0

Air System Sizing Summary for AC-05

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-05**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**42,5**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**18,9**	kW	Load occurs at	**Sep 1500**	
Sensible coil load	**14,6**	kW	OA DB / WB	**32,1 / 21,3**	°C
Coil L/s at Sep 1500	**1396**	L/s	Entering DB / WB	**23,1 / 17,2**	°C
Max block L/s	**1396**	L/s	Leaving DB / WB	**13,7 / 12,9**	°C
Sum of peak zone L/s	**1396**	L/s	Coil ADP	**12,0**	°C
Sensible heat ratio	**0,769**		Bypass Factor	**0,150**	
m²/kW	**2,2**		Resulting RH	**58**	%
W/m²	**445,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**1396**	L/s	Fan motor BHP	**0,98**	BHP
Standard L/s	**1290**	L/s	Fan motor kW	**0,78**	kW
Actual max L/(s-m²)	**32,84**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**54**	L/s	L/s/person	**7,73**	L/s/person
L/(s-m²)	**1,27**	L/(s-m²)			

Zone Sizing Summary for AC-05

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-05**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **42,5** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	13,7	1368	1368	Oct 1500	2,5	42,5	32,19

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Circulação 1	1	13,7	Oct 1500	1368	2,5	42,5	32,19

Air System Design Load Summary for AC-05

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Sep 1500 COOLING OA DB / WB 32,1 °C / 21,3 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	17 m²	2118	-	17 m²	-	-
Wall Transmission	23 m²	756	-	23 m²	597	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	17 m²	805	-	17 m²	1073	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	43 m²	1020	-	43 m²	561	-
Overhead Lighting	510 W	277	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	250 W	225	-	0	0	-
People	7	362	421	0	0	0
Infiltration	-	6595	3275	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	1216	370	10%	223	0
>> Total Zone Loads	**-**	**13375**	**4065**	**-**	**2454**	**0**
Zone Conditioning	-	12667	4065	-	157	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	153	-	0	0	-
Return Fan Load	1396 L/s	0	-	1396 L/s	0	-
Ventilation Load	54 L/s	567	306	54 L/s	516	0
Supply Fan Load	1396 L/s	775	-	1396 L/s	-775	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	401	-	3%	74	-
>> Total System Loads	**-**	**14564**	**4371**	**-**	**-29**	**0**
Central Cooling Coil	-	14564	4371	-	0	0
>> Total Conditioning	**-**	**14564**	**4371**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

System Psychrometrics for AC-05

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

September DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,1	0,01283	54	400	567	306
Vent - Return Mixing	Outlet	23,1	0,01084	1396	475	-	-
Central Cooling Coil	Outlet	13,7	0,00969	1396	475	14564	4371
Supply Fan	Outlet	14,2	0,00969	1396	475	775	-
Cold Supply Duct	Outlet	14,5	0,00969	1368	475	-	-
Zone Air	-	22,8	0,01078	1368	478	12667	4065
Return Plenum	Outlet	22,9	0,01078	1368	478	153	-
Duct Leakage Air	Outlet	14,2	0,00969	28	475	-	-
Return Duct	Outlet	22,7	0,01075	1396	478	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	13375	Cooling	12667	22,8	1368	478	0	0

Air System Sizing Summary for AC-06

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-06**
Equipment Class	**SPLT AHU**
Air System Type	**SZCAV**
Number of zones	**1**
Floor Area	**6,7** m²
Location	**Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**
Sizing Data	**Calculated**
Zone L/s Sizing	**Sum of space airflow rates**
Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1,2**	kW
Sensible coil load	**0,8**	kW
Coil L/s at Jul 1500	**53**	L/s
Max block L/s	**53**	L/s
Sum of peak zone L/s	**53**	L/s
Sensible heat ratio	**0,678**	
m²/kW	**5,4**	
W/m²	**184,8**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**33,2 / 21,9**	°C
Entering DB / WB	**27,3 / 19,3**	°C
Leaving DB / WB	**13,2 / 12,2**	°C
Coil ADP	**10,7**	°C
Bypass Factor	**0,150**	
Resulting RH	**59**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**53**	L/s
Standard L/s	**49**	L/s
Actual max L/(s-m²)	**7,95**	L/(s-m²)
Fan motor BHP	**0,04**	BHP
Fan motor kW	**0,03**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**23**	L/s
L/(s-m²)	**3,36**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-06

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-06**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **6,7** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,5	52	52	Jul 1500	0,1	6,7	7,79

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Descanso 1	1	0,5	Jul 1500	52	0,1	6,7	7,79

Air System Design Load Summary for AC-06

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	7 m²	165	-	7 m²	88	-
Overhead Lighting	80 W	44	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	100 W	90	-	0	0	-
People	3	178	237	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	48	24	10%	9	0
>> Total Zone Loads	**-**	**524**	**261**	**-**	**97**	**0**
Zone Conditioning	-	514	261	-	-36	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	24	-	0	0	-
Return Fan Load	53 L/s	0	-	53 L/s	0	-
Ventilation Load	23 L/s	256	138	23 L/s	63	0
Supply Fan Load	53 L/s	30	-	53 L/s	-30	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	16	-	3%	3	-
>> Total System Loads	**-**	**839**	**399**	**-**	**0**	**0**
Central Cooling Coil	-	839	399	-	0	0
>> Total Conditioning	**-**	**839**	**399**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-06

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	23	400	256	138
Vent - Return Mixing	Outlet	27,3	0,01190	53	924	-	-
Central Cooling Coil	Outlet	13,2	0,00915	53	924	839	399
Supply Fan	Outlet	13,7	0,00915	53	924	30	-
Cold Supply Duct	Outlet	14,0	0,00915	52	924	-	-
Zone Air	-	22,8	0,01099	52	1316	514	261
Return Plenum	Outlet	23,2	0,01099	52	1316	24	-
Duct Leakage Air	Outlet	13,7	0,00915	1	924	-	-
Return Duct	Outlet	23,0	0,01095	53	1308	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	524	Cooling	514	22,8	52	1316	0	0

Air System Sizing Summary for AC-07

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-07**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**6,2** m²
Air System Type	**SZCAV**	Location	**Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1,3**	kW	Load occurs at	**Jun 1500**	
Sensible coil load	**0,9**	kW	OA DB / WB	**32,6 / 21,9**	°C
Coil L/s at Jun 1500	**58**	L/s	Entering DB / WB	**26,7 / 18,9**	°C
Max block L/s	**58**	L/s	Leaving DB / WB	**13,2 / 12,2**	°C
Sum of peak zone L/s	**58**	L/s	Coil ADP	**10,8**	°C
Sensible heat ratio	**0,695**		Bypass Factor	**0,150**	
m²/kW	**4,9**		Resulting RH	**56**	%
W/m²	**202,8**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**58**	L/s	Fan motor BHP	**0,04**	BHP
Standard L/s	**54**	L/s	Fan motor kW	**0,03**	kW
Actual max L/(s-m²)	**9,36**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**23**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**3,63**	L/(s-m²)			

Zone Sizing Summary for AC-07

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-07**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **6,2** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,6	57	57	Jul 1500	0,2	6,2	9,18

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Descanso 2	1	0,6	Jul 1500	57	0,2	6,2	9,18

Air System Design Load Summary for AC-07

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jun 1500 COOLING OA DB / WB 32,6 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	6 m²	170	-	6 m²	91	-
Ceiling	6 m²	152	-	6 m²	82	-
Overhead Lighting	74 W	40	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	0 W	0	-	0	0	-
People	3	155	180	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	52	18	10%	17	0
>> Total Zone Loads	**-**	**570**	**198**	**-**	**191**	**0**
Zone Conditioning	-	558	198	-	-47	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	22	-	0	0	-
Return Fan Load	58 L/s	0	-	58 L/s	0	-
Ventilation Load	23 L/s	244	186	23 L/s	74	0
Supply Fan Load	58 L/s	32	-	58 L/s	-32	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	17	-	3%	6	-
>> Total System Loads	**-**	**874**	**384**	**-**	**1**	**0**
Central Cooling Coil	-	874	384	-	0	0
>> Total Conditioning	**-**	**874**	**384**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

June DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,6	0,01344	23	400	244	186
Vent - Return Mixing	Outlet	26,7	0,01158	58	848	-	-
Central Cooling Coil	Outlet	13,2	0,00916	58	848	874	384
Supply Fan	Outlet	13,7	0,00916	58	848	32	-
Cold Supply Duct	Outlet	14,0	0,00916	57	848	-	-
Zone Air	-	22,7	0,01044	57	1138	558	198
Return Plenum	Outlet	23,1	0,01044	57	1138	22	-
Duct Leakage Air	Outlet	13,7	0,00916	1	848	-	-
Return Duct	Outlet	22,9	0,01041	58	1132	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	570	Cooling	558	22,7	57	1138	0	0

Air System Sizing Summary for AC-08

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-08**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**118,0**	m²
Air System Type	**SZCAV**		Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**22,2**	kW	Load occurs at	**Nov 1400**	
Sensible coil load	**19,4**	kW	OA DB / WB	**29,0 / 20,2**	°C
Coil L/s at Nov 1400	**1721**	L/s	Entering DB / WB	**23,4 / 16,5**	°C
Max block L/s	**1721**	L/s	Leaving DB / WB	**13,3 / 12,3**	°C
Sum of peak zone L/s	**1721**	L/s	Coil ADP	**11,5**	°C
Sensible heat ratio	**0,872**		Bypass Factor	**0,150**	
m²/kW	**5,3**		Resulting RH	**51**	%
W/m²	**188,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**1721**	L/s	Fan motor BHP	**1,21**	BHP
Standard L/s	**1590**	L/s	Fan motor kW	**0,96**	kW
Actual max L/(s-m²)	**14,58**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**173**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**1,46**	L/(s-m²)			

Zone Sizing Summary for AC-08

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-08**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **118,0** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	16,9	1686	1686	Dec 1500	5,2	118,0	14,29

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Hall 1	1	16,9	Dec 1500	1686	5,2	118,0	14,29

Air System Design Load Summary for AC-08

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Nov 1400 COOLING OA DB / WB 29,0 °C / 20,2 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	32 m²	7913	-	32 m²	-	-
Wall Transmission	43 m²	1563	-	43 m²	1124	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	32 m²	946	-	32 m²	2007	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	118 m²	2206	-	118 m²	1558	-
Overhead Lighting	1416 W	754	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	668 W	597	-	0	0	-
People	23	1159	1382	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	1514	138	10%	469	0
>> Total Zone Loads	**-**	**16651**	**1521**	**-**	**5157**	**0**
Zone Conditioning	-	16310	1521	-	304	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	425	-	0	0	-
Return Fan Load	1721 L/s	0	-	1721 L/s	0	-
Ventilation Load	173 L/s	1207	1316	173 L/s	453	0
Supply Fan Load	1721 L/s	956	-	1721 L/s	-956	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	500	-	3%	155	-
>> Total System Loads	**-**	**19397**	**2837**	**-**	**-45**	**0**
Central Cooling Coil	-	19397	2841	-	0	0
>> Total Conditioning	**-**	**19397**	**2841**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

November DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	29,0	0,01237	173	400	1207	1316
Vent - Return Mixing	Outlet	23,4	0,00985	1721	992	-	-
Central Cooling Coil	Outlet	13,3	0,00924	1721	992	19397	2841
Supply Fan	Outlet	13,8	0,00924	1721	992	956	-
Cold Supply Duct	Outlet	14,0	0,00924	1686	992	-	-
Zone Air	-	22,7	0,00957	1686	1059	16310	1521
Return Plenum	Outlet	22,9	0,00957	1686	1059	425	-
Duct Leakage Air	Outlet	13,8	0,00924	34	992	-	-
Return Duct	Outlet	22,7	0,00956	1721	1058	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	16651	Cooling	16310	22,7	1686	1059	0	0

Air System Sizing Summary for AC-09

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-09**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**109,0**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**18,9**	kW	Load occurs at	**Nov 1400**	
Sensible coil load	**16,6**	kW	OA DB / WB	**29,0 / 20,2**	°C
Coil L/s at Nov 1400	**1513**	L/s	Entering DB / WB	**23,1 / 16,4**	°C
Max block L/s	**1513**	L/s	Leaving DB / WB	**13,3 / 12,4**	°C
Sum of peak zone L/s	**1513**	L/s	Coil ADP	**11,6**	°C
Sensible heat ratio	**0,879**		Bypass Factor	**0,150**	
m²/kW	**5,8**		Resulting RH	**52**	%
W/m²	**173,3**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**1513**	L/s	Fan motor BHP	**1,06**	BHP
Standard L/s	**1398**	L/s	Fan motor kW	**0,84**	kW
Actual max L/(s-m²)	**13,88**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**83**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**0,76**	L/(s-m²)			

Zone Sizing Summary for AC-09

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-09**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **109,0** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	14,9	1483	1483	Dec 1500	3,6	109,0	13,61

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Recepção 1	1	14,9	Dec 1500	1483	3,6	109,0	13,61

Air System Design Load Summary for AC-09

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Nov 1400 COOLING OA DB / WB 29,0 °C / 20,2 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	34 m²	8281	-	34 m²	-	-
Wall Transmission	44 m²	1625	-	44 m²	1169	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	34 m²	990	-	34 m²	2100	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	1308 W	697	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	256 W	229	-	0	0	-
People	11	554	661	0	0	0
Infiltration	-	939	872	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	1332	153	10%	327	0
>> Total Zone Loads	**-**	**14647**	**1686**	**-**	**3596**	**0**
Zone Conditioning	-	14355	1686	-	356	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	392	-	0	0	-
Return Fan Load	1513 L/s	0	-	1513 L/s	0	-
Ventilation Load	83 L/s	572	601	83 L/s	352	0
Supply Fan Load	1513 L/s	841	-	1513 L/s	-841	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	439	-	3%	108	-
>> Total System Loads	**-**	**16600**	**2287**	**-**	**-24**	**0**
Central Cooling Coil	-	16600	2291	-	0	0
>> Total Conditioning	**-**	**16600**	**2291**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

November DESIGN COOLING DAY, 1400

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	29,0	0,01237	83	400	572	601
Vent - Return Mixing	Outlet	23,1	0,00984	1513	755	-	-
Central Cooling Coil	Outlet	13,3	0,00928	1513	755	16600	2291
Supply Fan	Outlet	13,8	0,00928	1513	755	841	-
Cold Supply Duct	Outlet	14,1	0,00928	1483	755	-	-
Zone Air	-	22,7	0,00970	1483	776	14355	1686
Return Plenum	Outlet	23,0	0,00970	1483	776	392	-
Duct Leakage Air	Outlet	13,8	0,00928	30	755	-	-
Return Duct	Outlet	22,8	0,00969	1513	776	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	14647	Cooling	14355	22,7	1483	776	0	0

Air System Sizing Summary for AC-10

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-10**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**122,6**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**23,4**	kW	Load occurs at	**Jun 1500**	
Sensible coil load	**16,6**	kW	OA DB / WB	**32,6 / 21,9**	°C
Coil L/s at Jun 1500	**1063**	L/s	Entering DB / WB	**27,1 / 18,8**	°C
Max block L/s	**1063**	L/s	Leaving DB / WB	**13,0 / 12,0**	°C
Sum of peak zone L/s	**1063**	L/s	Coil ADP	**10,6**	°C
Sensible heat ratio	**0,710**		Bypass Factor	**0,150**	
m²/kW	**5,2**		Resulting RH	**53**	%
W/m²	**190,7**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**1063**	L/s	Fan motor BHP	**0,74**	BHP
Standard L/s	**982**	L/s	Fan motor kW	**0,59**	kW
Actual max L/(s-m²)	**8,67**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**450**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**3,67**	L/(s-m²)			

Zone Sizing Summary for AC-10

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-10**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **122,6** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	10,5	1041	1041	Jul 1500	1,5	122,6	8,49

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Refeitório 1	1	10,5	Jul 1500	1041	1,5	122,6	8,49

Air System Design Load Summary for AC-10

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jun 1500 COOLING OA DB / WB 32,6 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	92 m²	2622	-	92 m²	1408	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	1471 W	800	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	3510 W	3160	-	0	0	-
People	60	2916	2112	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	950	211	10%	141	0
>> Total Zone Loads	**-**	**10448**	**2323**	**-**	**1549**	**0**
Zone Conditioning	-	10392	2323	-	-622	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	441	-	0	0	-
Return Fan Load	1063 L/s	0	-	1063 L/s	0	-
Ventilation Load	450 L/s	4867	4453	450 L/s	1175	0
Supply Fan Load	1063 L/s	590	-	1063 L/s	-590	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	313	-	3%	46	-
>> Total System Loads	**-**	**16605**	**6776**	**-**	**9**	**0**
Central Cooling Coil	-	16605	6777	-	0	0
>> Total Conditioning	**-**	**16605**	**6777**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-10

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

June DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,6	0,01344	450	400	4867	4453
Vent - Return Mixing	Outlet	27,1	0,01134	1063	750	-	-
Central Cooling Coil	Outlet	13,0	0,00900	1063	750	16605	6777
Supply Fan	Outlet	13,5	0,00900	1063	750	590	-
Cold Supply Duct	Outlet	13,8	0,00900	1041	750	-	-
Zone Air	-	22,8	0,00982	1041	1013	10392	2323
Return Plenum	Outlet	23,1	0,00982	1041	1013	441	-
Duct Leakage Air	Outlet	13,5	0,00900	21	750	-	-
Return Duct	Outlet	22,9	0,00980	1063	1008	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	10448	Cooling	10392	22,8	1041	1013	0	0

Air System Sizing Summary for AC-11

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-11**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**11,1** m²
Air System Type	**SZCAV**	Location	**Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,4**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**1,9**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**146**	L/s	Entering DB / WB	**25,1 / 17,7**	°C
Max block L/s	**146**	L/s	Leaving DB / WB	**13,4 / 12,4**	°C
Sum of peak zone L/s	**146**	L/s	Coil ADP	**11,4**	°C
Sensible heat ratio	**0,782**		Bypass Factor	**0,150**	
m²/kW	**4,6**		Resulting RH	**53**	%
W/m²	**218,8**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**146**	L/s	Fan motor BHP	**0,10**	BHP
Standard L/s	**135**	L/s	Fan motor kW	**0,08**	kW
Actual max L/(s-m²)	**13,17**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,70**	L/(s-m²)			

Zone Sizing Summary for AC-11

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-11**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **11,1** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,4	143	143	Jul 1500	0,4	11,1	12,91

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 10	1	1,4	Jul 1500	143	0,4	11,1	12,91

Air System Design Load Summary for AC-11

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	12 m²	339	-	12 m²	181	-
Ceiling	11 m²	273	-	11 m²	147	-
Overhead Lighting	133 W	72	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	462 W	416	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	131	24	10%	33	0
>> Total Zone Loads	**-**	**1438**	**264**	**-**	**361**	**0**
Zone Conditioning	-	1391	264	-	-51	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	40	-	0	0	-
Return Fan Load	146 L/s	0	-	146 L/s	0	-
Ventilation Load	30 L/s	343	266	30 L/s	123	0
Supply Fan Load	146 L/s	81	-	146 L/s	-81	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	43	-	3%	11	-
>> Total System Loads	**-**	**1898**	**530**	**-**	**2**	**0**
Central Cooling Coil	-	1898	530	-	0	0
>> Total Conditioning	**-**	**1898**	**530**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-11

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	30	400	343	266
Vent - Return Mixing	Outlet	25,1	0,01062	146	968	-	-
Central Cooling Coil	Outlet	13,4	0,00929	146	968	1898	530
Supply Fan	Outlet	13,9	0,00929	146	968	81	-
Cold Supply Duct	Outlet	14,2	0,00929	143	968	-	-
Zone Air	-	22,9	0,00997	143	1118	1391	264
Return Plenum	Outlet	23,1	0,00997	143	1118	40	-
Duct Leakage Air	Outlet	13,9	0,00929	3	968	-	-
Return Duct	Outlet	22,9	0,00995	146	1115	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1438	Cooling	1391	22,9	143	1118	0	0

Air System Sizing Summary for AC-12

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-12**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**10,2**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1,7**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**1,2**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**77**	L/s	Entering DB / WB	**27,1 / 19,0**	°C
Max block L/s	**77**	L/s	Leaving DB / WB	**13,5 / 12,4**	°C
Sum of peak zone L/s	**77**	L/s	Coil ADP	**11,1**	°C
Sensible heat ratio	**0,709**		Bypass Factor	**0,150**	
m²/kW	**6,2**		Resulting RH	**56**	%
W/m²	**162,2**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**77**	L/s	Fan motor BHP	**0,05**	BHP
Standard L/s	**71**	L/s	Fan motor kW	**0,04**	kW
Actual max L/(s-m²)	**7,56**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,94**	L/(s-m²)			

Zone Sizing Summary for AC-12

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-12**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **10,2** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,8	76	76	Jan 1500	0,0	10,2	7,41

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 2	1	0,8	Jan 1500	76	0,0	10,2	7,41

Air System Design Load Summary for AC-12

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	122 W	67	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	462 W	416	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	69	24	10%	0	0
>> Total Zone Loads	**-**	**759**	**264**	**-**	**0**	**0**
Zone Conditioning	-	737	264	-	-23	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	37	-	0	0	-
Return Fan Load	77 L/s	0	-	77 L/s	0	-
Ventilation Load	30 L/s	333	217	30 L/s	68	0
Supply Fan Load	77 L/s	43	-	77 L/s	-43	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	23	-	3%	0	-
>> Total System Loads	**-**	**1173**	**482**	**-**	**2**	**0**
Central Cooling Coil	-	1173	482	-	0	0
>> Total Conditioning	**-**	**1173**	**482**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	30	400	333	217
Vent - Return Mixing	Outlet	27,1	0,01158	77	847	-	-
Central Cooling Coil	Outlet	13,5	0,00929	77	847	1173	482
Supply Fan	Outlet	14,0	0,00929	77	847	43	-
Cold Supply Duct	Outlet	14,3	0,00929	76	847	-	-
Zone Air	-	23,0	0,01057	76	1138	737	264
Return Plenum	Outlet	23,4	0,01057	76	1138	37	-
Duct Leakage Air	Outlet	14,0	0,00929	2	847	-	-
Return Duct	Outlet	23,2	0,01055	77	1132	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	759	Cooling	737	23,0	76	1138	0	0

Air System Sizing Summary for AC-13

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-13**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**8,3** m²
Air System Type	**SZCAV**	Location	**Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1,7** kW	Load occurs at	**Jul 1500**	
Sensible coil load	**1,2** kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**76** L/s	Entering DB / WB	**27,1 / 19,0**	°C
Max block L/s	**76** L/s	Leaving DB / WB	**13,2 / 12,2**	°C
Sum of peak zone L/s	**76** L/s	Coil ADP	**10,8**	°C
Sensible heat ratio	**0,704**	Bypass Factor	**0,150**	
m²/kW	**5,0**	Resulting RH	**55**	%
W/m²	**200,3**	Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**	Zone T-stat Check	**1 of 1**	OK
		Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**76** L/s	Fan motor BHP	**0,05**	BHP
Standard L/s	**70** L/s	Fan motor kW	**0,04**	kW
Actual max L/(s-m²)	**9,12** L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30** L/s	L/s/person	**7,50** L/s/person
L/(s-m²)	**3,61** L/(s-m²)		

Zone Sizing Summary for AC-13

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-13**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **8,3** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	0,7	74	74	Jan 1500	0,0	8,3	8,94

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 3	1	0,7	Jan 1500	74	0,0	8,3	8,94

Air System Design Load Summary for AC-13

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	100 W	54	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	462 W	416	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	68	24	10%	0	0
>> Total Zone Loads	**-**	**745**	**264**	**-**	**0**	**0**
Zone Conditioning	-	737	264	-	-15	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	30	-	0	0	-
Return Fan Load	76 L/s	0	-	76 L/s	0	-
Ventilation Load	30 L/s	339	228	30 L/s	59	0
Supply Fan Load	76 L/s	42	-	76 L/s	-42	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	22	-	3%	0	-
>> Total System Loads	**-**	**1170**	**492**	**-**	**1**	**0**
Central Cooling Coil	-	1170	492	-	0	0
>> Total Conditioning	**-**	**1170**	**492**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	30	400	339	228
Vent - Return Mixing	Outlet	27,1	0,01152	76	844	-	-
Central Cooling Coil	Outlet	13,2	0,00913	76	844	1170	492
Supply Fan	Outlet	13,7	0,00913	76	844	42	-
Cold Supply Duct	Outlet	14,0	0,00913	74	844	-	-
Zone Air	-	22,9	0,01044	74	1141	737	264
Return Plenum	Outlet	23,3	0,01044	74	1141	30	-
Duct Leakage Air	Outlet	13,7	0,00913	2	844	-	-
Return Duct	Outlet	23,1	0,01042	76	1135	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	745	Cooling	737	22,9	74	1141	0	0

Air System Sizing Summary for AC-14

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-14**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**65,3**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**11,4**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**8,6**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**627**	L/s	Entering DB / WB	**25,4 / 17,7**	°C
Max block L/s	**627**	L/s	Leaving DB / WB	**13,0 / 12,0**	°C
Sum of peak zone L/s	**627**	L/s	Coil ADP	**10,8**	°C
Sensible heat ratio	**0,761**		Bypass Factor	**0,150**	
m²/kW	**5,8**		Resulting RH	**53**	%
W/m²	**173,9**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**627**	L/s	Fan motor BHP	**0,44**	BHP
Standard L/s	**579**	L/s	Fan motor kW	**0,35**	kW
Actual max L/(s-m²)	**9,60**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**150**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,30**	L/(s-m²)			

Zone Sizing Summary for AC-14

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-14**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **65,3** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	6,2	614	614	Jul 1500	1,3	65,3	9,41

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 4	1	6,2	Jul 1500	614	1,3	65,3	9,41

Air System Design Load Summary for AC-14

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	22 m²	637	-	22 m²	341	-
Ceiling	65 m²	1609	-	65 m²	862	-
Overhead Lighting	784 W	426	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	2110 W	1900	-	0	0	-
People	20	1036	1202	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	561	120	10%	120	0
>> Total Zone Loads	**-**	**6168**	**1322**	**-**	**1324**	**0**
Zone Conditioning	-	6144	1322	-	-265	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	235	-	0	0	-
Return Fan Load	627 L/s	0	-	627 L/s	0	-
Ventilation Load	150 L/s	1725	1394	150 L/s	582	0
Supply Fan Load	627 L/s	348	-	627 L/s	-348	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	185	-	3%	40	-
>> Total System Loads	**-**	**8637**	**2717**	**-**	**8**	**0**
Central Cooling Coil	-	8637	2718	-	0	0
>> Total Conditioning	**-**	**8637**	**2718**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-14

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	150	400	1725	1394
Vent - Return Mixing	Outlet	25,4	0,01061	627	949	-	-
Central Cooling Coil	Outlet	13,0	0,00902	627	949	8637	2718
Supply Fan	Outlet	13,5	0,00902	627	949	348	-
Cold Supply Duct	Outlet	13,8	0,00902	614	949	-	-
Zone Air	-	22,7	0,00981	614	1125	6144	1322
Return Plenum	Outlet	23,1	0,00981	614	1125	235	-
Duct Leakage Air	Outlet	13,5	0,00902	13	949	-	-
Return Duct	Outlet	22,9	0,00979	627	1121	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	6168	Cooling	6144	22,7	614	1125	0	0

Air System Sizing Summary for AC-15

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-15**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**55,1**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**9,6**	kW	Load occurs at	**Jun 1500**	
Sensible coil load	**7,3**	kW	OA DB / WB	**32,6 / 21,9**	°C
Coil L/s at Jun 1500	**546**	L/s	Entering DB / WB	**25,0 / 17,6**	°C
Max block L/s	**546**	L/s	Leaving DB / WB	**13,0 / 12,0**	°C
Sum of peak zone L/s	**546**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,764**		Bypass Factor	**0,150**	
m²/kW	**5,8**		Resulting RH	**52**	%
W/m²	**173,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**546**	L/s	Fan motor BHP	**0,38**	BHP
Standard L/s	**505**	L/s	Fan motor kW	**0,30**	kW
Actual max L/(s-m²)	**9,91**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**120**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,18**	L/(s-m²)			

Zone Sizing Summary for AC-15

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-15**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **55,1** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	5,4	535	535	Jul 1500	1,3	55,1	9,71

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 5	1	5,4	Jul 1500	535	1,3	55,1	9,71

Air System Design Load Summary for AC-15

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jun 1500 COOLING OA DB / WB 32,6 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	30 m²	854	-	30 m²	459	-
Ceiling	55 m²	1354	-	55 m²	727	-
Overhead Lighting	661 W	359	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1648 W	1484	-	0	0	-
People	16	828	962	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	488	96	10%	119	0
>> Total Zone Loads	**-**	**5368**	**1058**	**-**	**1305**	**0**
Zone Conditioning	-	5330	1058	-	-216	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	198	-	0	0	-
Return Fan Load	546 L/s	0	-	546 L/s	0	-
Ventilation Load	120 L/s	1308	1198	120 L/s	487	0
Supply Fan Load	546 L/s	303	-	546 L/s	-303	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	161	-	3%	39	-
>> Total System Loads	**-**	**7301**	**2256**	**-**	**7**	**0**
Central Cooling Coil	-	7301	2257	-	0	0
>> Total Conditioning	**-**	**7301**	**2257**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-15

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

June DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	32,6	0,01344	120	400	1308	1198
Vent - Return Mixing	Outlet	25,0	0,01058	546	956	-	-
Central Cooling Coil	Outlet	13,0	0,00906	546	956	7301	2257
Supply Fan	Outlet	13,5	0,00906	546	956	303	-
Cold Supply Duct	Outlet	13,8	0,00906	535	956	-	-
Zone Air	-	22,7	0,00979	535	1116	5330	1058
Return Plenum	Outlet	23,1	0,00979	535	1116	198	-
Duct Leakage Air	Outlet	13,5	0,00906	11	956	-	-
Return Duct	Outlet	22,9	0,00977	546	1112	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	5368	Cooling	5330	22,7	535	1116	0	0

Air System Sizing Summary for AC-16

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-16**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**65,3**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**10,8**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**8,1**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**580**	L/s	Entering DB / WB	**25,6 / 17,9**	°C
Max block L/s	**580**	L/s	Leaving DB / WB	**13,0 / 12,0**	°C
Sum of peak zone L/s	**580**	L/s	Coil ADP	**10,8**	°C
Sensible heat ratio	**0,751**		Bypass Factor	**0,150**	
m²/kW	**6,0**		Resulting RH	**53**	%
W/m²	**165,3**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**580**	L/s	Fan motor BHP	**0,41**	BHP
Standard L/s	**536**	L/s	Fan motor kW	**0,32**	kW
Actual max L/(s-m²)	**8,89**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**150**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,30**	L/(s-m²)			

Zone Sizing Summary for AC-16

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-16**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **65,3** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	5,7	569	569	Jul 1500	1,3	65,3	8,71

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 6	1	5,7	Jul 1500	569	1,3	65,3	8,71

Air System Design Load Summary for AC-16

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	22 m²	637	-	22 m²	341	-
Ceiling	65 m²	1609	-	65 m²	862	-
Overhead Lighting	784 W	426	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1648 W	1484	-	0	0	-
People	20	1036	1202	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	519	120	10%	120	0
>> Total Zone Loads	**-**	**5711**	**1322**	**-**	**1324**	**0**
Zone Conditioning	-	5661	1322	-	-280	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	235	-	0	0	-
Return Fan Load	580 L/s	0	-	580 L/s	0	-
Ventilation Load	150 L/s	1721	1360	150 L/s	570	0
Supply Fan Load	580 L/s	322	-	580 L/s	-322	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	171	-	3%	40	-
>> Total System Loads	**-**	**8111**	**2682**	**-**	**7**	**0**
Central Cooling Coil	-	8111	2683	-	0	0
>> Total Conditioning	**-**	**8111**	**2683**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-16

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	150	400	1721	1360
Vent - Return Mixing	Outlet	25,6	0,01074	580	933	-	-
Central Cooling Coil	Outlet	13,0	0,00904	580	933	8111	2683
Supply Fan	Outlet	13,5	0,00904	580	933	322	-
Cold Supply Duct	Outlet	13,8	0,00904	569	933	-	-
Zone Air	-	22,7	0,00989	569	1123	5661	1322
Return Plenum	Outlet	23,1	0,00989	569	1123	235	-
Duct Leakage Air	Outlet	13,5	0,00904	12	933	-	-
Return Duct	Outlet	22,9	0,00988	580	1119	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	5711	Cooling	5661	22,7	569	1123	0	0

Air System Sizing Summary for AC-17

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-17**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**15,8** m²
Air System Type	**SZCAV**	Location	**Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3,6**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**2,8**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**214**	L/s	Entering DB / WB	**25,1 / 17,7**	°C
Max block L/s	**214**	L/s	Leaving DB / WB	**13,4 / 12,4**	°C
Sum of peak zone L/s	**214**	L/s	Coil ADP	**11,3**	°C
Sensible heat ratio	**0,778**		Bypass Factor	**0,150**	
m²/kW	**4,4**		Resulting RH	**53**	%
W/m²	**227,0**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**214**	L/s	Fan motor BHP	**0,15**	BHP
Standard L/s	**198**	L/s	Fan motor kW	**0,12**	kW
Actual max L/(s-m²)	**13,54**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**45**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,85**	L/(s-m²)			

Zone Sizing Summary for AC-17

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-17**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **15,8** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	2,1	210	210	Jul 1500	0,5	15,8	13,27

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 7	1	2,1	Jul 1500	210	0,5	15,8	13,27

Air System Design Load Summary for AC-17

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	18 m²	509	-	18 m²	273	-
Ceiling	16 m²	389	-	16 m²	209	-
Overhead Lighting	190 W	103	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	668 W	601	-	0	0	-
People	6	311	361	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	191	36	10%	48	0
>> Total Zone Loads	**-**	**2105**	**397**	**-**	**529**	**0**
Zone Conditioning	-	2039	397	-	-77	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	57	-	0	0	-
Return Fan Load	214 L/s	0	-	214 L/s	0	-
Ventilation Load	45 L/s	515	398	45 L/s	183	0
Supply Fan Load	214 L/s	119	-	214 L/s	-119	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	63	-	3%	16	-
>> Total System Loads	**-**	**2793**	**795**	**-**	**3**	**0**
Central Cooling Coil	-	2793	795	-	0	0
>> Total Conditioning	**-**	**2793**	**795**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-17

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	45	400	515	398
Vent - Return Mixing	Outlet	25,1	0,01064	214	967	-	-
Central Cooling Coil	Outlet	13,4	0,00928	214	967	2793	795
Supply Fan	Outlet	13,9	0,00928	214	967	119	-
Cold Supply Duct	Outlet	14,2	0,00928	210	967	-	-
Zone Air	-	22,9	0,00997	210	1121	2039	397
Return Plenum	Outlet	23,1	0,00997	210	1121	57	-
Duct Leakage Air	Outlet	13,9	0,00928	4	967	-	-
Return Duct	Outlet	22,9	0,00996	214	1117	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	2105	Cooling	2039	22,9	210	1121	0	0

Air System Sizing Summary for AC-18

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-18**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**12,4**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,1**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**1,6**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**113**	L/s	Entering DB / WB	**25,7 / 18,0**	°C
Max block L/s	**113**	L/s	Leaving DB / WB	**13,1 / 12,1**	°C
Sum of peak zone L/s	**113**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,748**		Bypass Factor	**0,150**	
m²/kW	**5,9**		Resulting RH	**53**	%
W/m²	**170,7**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**113**	L/s	Fan motor BHP	**0,08**	BHP
Standard L/s	**104**	L/s	Fan motor kW	**0,06**	kW
Actual max L/(s-m²)	**9,10**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**2,42**	L/(s-m²)			

Zone Sizing Summary for AC-18

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-18**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **12,4** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,1	111	111	Jul 1500	0,2	12,4	8,92

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 8	1	1,1	Jul 1500	111	0,2	12,4	8,92

Air System Design Load Summary for AC-18

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	12 m²	306	-	12 m²	164	-
Overhead Lighting	149 W	81	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	462 W	416	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	101	24	10%	16	0
>> Total Zone Loads	**-**	**1110**	**264**	**-**	**180**	**0**
Zone Conditioning	-	1099	264	-	-49	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	45	-	0	0	-
Return Fan Load	113 L/s	0	-	113 L/s	0	-
Ventilation Load	30 L/s	344	268	30 L/s	108	0
Supply Fan Load	113 L/s	63	-	113 L/s	-63	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	33	-	3%	5	-
>> Total System Loads	**-**	**1584**	**533**	**-**	**1**	**0**
Central Cooling Coil	-	1584	533	-	0	0
>> Total Conditioning	**-**	**1584**	**533**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-18

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	30	400	344	268
Vent - Return Mixing	Outlet	25,7	0,01080	113	930	-	-
Central Cooling Coil	Outlet	13,1	0,00906	113	930	1584	533
Supply Fan	Outlet	13,6	0,00906	113	930	63	-
Cold Supply Duct	Outlet	13,8	0,00906	111	930	-	-
Zone Air	-	22,8	0,00994	111	1126	1099	264
Return Plenum	Outlet	23,1	0,00994	111	1126	45	-
Duct Leakage Air	Outlet	13,6	0,00906	2	930	-	-
Return Duct	Outlet	22,9	0,00992	113	1122	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1110	Cooling	1099	22,8	111	1126	0	0

Air System Sizing Summary for AC-19

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-19**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**14,5**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3,4**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**2,6**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**190**	L/s	Entering DB / WB	**25,2 / 17,7**	°C
Max block L/s	**190**	L/s	Leaving DB / WB	**13,1 / 12,0**	°C
Sum of peak zone L/s	**190**	L/s	Coil ADP	**10,9**	°C
Sensible heat ratio	**0,761**		Bypass Factor	**0,150**	
m²/kW	**4,3**		Resulting RH	**53**	%
W/m²	**234,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**190**	L/s	Fan motor BHP	**0,13**	BHP
Standard L/s	**176**	L/s	Fan motor kW	**0,11**	kW
Actual max L/(s-m²)	**13,13**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**45**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**3,10**	L/(s-m²)			

Zone Sizing Summary for AC-19

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-19**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **14,5** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,9	187	187	Jul 1500	0,4	14,5	12,86

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 9	1	1,9	Jul 1500	187	0,4	14,5	12,86

Air System Design Load Summary for AC-19

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	12 m²	339	-	12 m²	181	-
Ceiling	15 m²	357	-	15 m²	191	-
Overhead Lighting	174 W	95	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	668 W	601	-	0	0	-
People	6	311	361	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	170	36	10%	37	0
>> Total Zone Loads	**-**	**1873**	**397**	**-**	**410**	**0**
Zone Conditioning	-	1851	397	-	-75	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	52	-	0	0	-
Return Fan Load	190 L/s	0	-	190 L/s	0	-
Ventilation Load	45 L/s	523	415	45 L/s	172	0
Supply Fan Load	190 L/s	106	-	190 L/s	-106	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	56	-	3%	12	-
>> Total System Loads	**-**	**2588**	**811**	**-**	**3**	**0**
Central Cooling Coil	-	2588	811	-	0	0
>> Total Conditioning	**-**	**2588**	**811**	**-**	**0**	**0**
Key:	**Positive values are clg loads Negative values are htg loads**			**Positive values are htg loads Negative values are clg loads**		

System Psychrometrics for AC-19

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	45	400	523	415
Vent - Return Mixing	Outlet	25,2	0,01062	190	950	-	-
Central Cooling Coil	Outlet	13,1	0,00906	190	950	2588	811
Supply Fan	Outlet	13,6	0,00906	190	950	106	-
Cold Supply Duct	Outlet	13,8	0,00906	187	950	-	-
Zone Air	-	22,7	0,00984	187	1124	1851	397
Return Plenum	Outlet	23,0	0,00984	187	1124	52	-
Duct Leakage Air	Outlet	13,6	0,00906	4	950	-	-
Return Duct	Outlet	22,8	0,00982	190	1120	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1873	Cooling	1851	22,7	187	1124	0	0

Air System Sizing Summary for AC-20

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-20**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**14,8**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**		Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**		Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2,6**	kW	Load occurs at	**Jul 1500**	
Sensible coil load	**1,9**	kW	OA DB / WB	**33,2 / 21,9**	°C
Coil L/s at Jul 1500	**121**	L/s	Entering DB / WB	**27,2 / 18,8**	°C
Max block L/s	**121**	L/s	Leaving DB / WB	**13,1 / 12,0**	°C
Sum of peak zone L/s	**121**	L/s	Coil ADP	**10,6**	°C
Sensible heat ratio	**0,724**		Bypass Factor	**0,150**	
m²/kW	**5,6**		Resulting RH	**53**	%
W/m²	**177,6**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**121**	L/s	Fan motor BHP	**0,08**	BHP
Standard L/s	**112**	L/s	Fan motor kW	**0,07**	kW
Actual max L/(s-m²)	**8,19**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**50**	L/s	L/s/person	**12,40**	L/s/person
L/(s-m²)	**3,35**	L/(s-m²)			

Zone Sizing Summary for AC-20

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-20**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **14,8** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	1,2	119	119	Jul 1500	0,2	14,8	8,03

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala geral 1	1	1,2	Jul 1500	119	0,2	14,8	8,03

Air System Design Load Summary for AC-20

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	15 m²	365	-	15 m²	195	-
Overhead Lighting	178 W	97	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	462 W	416	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	108	24	10%	20	0
>> Total Zone Loads	**-**	**1193**	**264**	**-**	**215**	**0**
Zone Conditioning	-	1181	264	-	-78	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	53	-	0	0	-
Return Fan Load	121 L/s	0	-	121 L/s	0	-
Ventilation Load	50 L/s	565	461	50 L/s	140	0
Supply Fan Load	121 L/s	67	-	121 L/s	-67	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	36	-	3%	6	-
>> Total System Loads	**-**	**1903**	**725**	**-**	**1**	**0**
Central Cooling Coil	-	1903	725	-	0	0
>> Total Conditioning	**-**	**1903**	**725**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-20

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	50	400	565	461
Vent - Return Mixing	Outlet	27,2	0,01119	121	679	-	-
Central Cooling Coil	Outlet	13,1	0,00900	121	679	1903	725
Supply Fan	Outlet	13,6	0,00900	121	679	67	-
Cold Supply Duct	Outlet	13,9	0,00900	119	679	-	-
Zone Air	-	22,8	0,00981	119	876	1181	264
Return Plenum	Outlet	23,2	0,00981	119	876	53	-
Duct Leakage Air	Outlet	13,6	0,00900	2	679	-	-
Return Duct	Outlet	23,0	0,00980	121	873	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	1193	Cooling	1181	22,8	119	876	0	0

Air System Sizing Summary for AC-21

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-21**				
Equipment Class	**SPLT AHU**		Number of zones	**1**	
Air System Type	**SZCAV**		Floor Area	**514,4**	m²
			Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone L/s Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space L/s Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**74,8**	kW	Load occurs at	**Oct 1500**	
Sensible coil load	**57,3**	kW	OA DB / WB	**31,0 / 20,8**	°C
Coil L/s at Oct 1500	**4445**	L/s	Entering DB / WB	**24,7 / 17,5**	°C
Max block L/s	**4445**	L/s	Leaving DB / WB	**13,1 / 12,2**	°C
Sum of peak zone L/s	**4445**	L/s	Coil ADP	**11,1**	°C
Sensible heat ratio	**0,766**		Bypass Factor	**0,150**	
m²/kW	**6,9**		Resulting RH	**54**	%
W/m²	**145,5**		Design supply temp.	**13,0**	°C
Water flow @ 5,6 °K rise	**N/A**		Zone T-stat Check	**1 of 1**	OK
			Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**4445**	L/s	Fan motor BHP	**3,11**	BHP
Standard L/s	**4107**	L/s	Fan motor kW	**2,47**	kW
Actual max L/(s-m²)	**8,64**	L/(s-m²)	Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**975**	L/s	L/s/person	**7,50**	L/s/person
L/(s-m²)	**1,90**	L/(s-m²)			

Zone Sizing Summary for AC-21

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-21**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **514,4** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	43,7	4356	4356	Nov 1500	4,9	514,4	8,47

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Workplace 1	1	43,7	Nov 1500	4356	4,9	514,4	8,47

Air System Design Load Summary for AC-21

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Oct 1500 COOLING OA DB / WB 31,0 °C / 20,8 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	32 m²	5899	-	32 m²	-	-
Wall Transmission	46 m²	1825	-	46 m²	1217	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	32 m²	1274	-	32 m²	1960	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	83 m²	2094	-	83 m²	1265	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	6173 W	3356	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	16270 W	14649	-	0	0	-
People	130	6731	7813	0	0	0
Infiltration	-	3535	2293	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	3936	1011	10%	444	0
>> Total Zone Loads	**-**	**43300**	**11117**	**-**	**4887**	**0**
Zone Conditioning	-	42952	11117	-	-2089	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	1852	-	0	0	-
Return Fan Load	4445 L/s	0	-	4445 L/s	0	-
Ventilation Load	975 L/s	8757	6383	975 L/s	4477	0
Supply Fan Load	4445 L/s	2469	-	4445 L/s	-2469	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	1299	-	3%	147	-
>> Total System Loads	**-**	**57329**	**17499**	**-**	**65**	**0**
Central Cooling Coil	-	57329	17505	-	0	0
>> Total Conditioning	**-**	**57329**	**17505**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

October DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	31,0	0,01247	975	400	8757	6383
Vent - Return Mixing	Outlet	24,7	0,01059	4445	809	-	-
Central Cooling Coil	Outlet	13,1	0,00915	4445	809	57329	17505
Supply Fan	Outlet	13,6	0,00915	4445	809	2469	-
Cold Supply Duct	Outlet	13,9	0,00915	4356	809	-	-
Zone Air	-	22,7	0,01008	4356	926	42952	11117
Return Plenum	Outlet	23,1	0,01008	4356	926	1852	-
Duct Leakage Air	Outlet	13,6	0,00915	89	809	-	-
Return Duct	Outlet	22,9	0,01007	4445	924	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	43300	Cooling	42952	22,7	4356	926	0	0

Air System Sizing Summary for AC-22

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-22**	
Equipment Class	**SPLT AHU**	
Air System Type	**SZCAV**	
Number of zones	**1**	
Floor Area	**10,4**	m²
Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	
Sizing Data	**Calculated**	
Zone L/s Sizing	**Sum of space airflow rates**	
Space L/s Sizing	**Individual peak space loads**	

Central Cooling Coil Sizing Data

Total coil load	**2,9**	kW
Sensible coil load	**2,4**	kW
Coil L/s at Aug 1800	**206**	L/s
Max block L/s	**206**	L/s
Sum of peak zone L/s	**206**	L/s
Sensible heat ratio	**0,831**	
m²/kW	**3,6**	
W/m²	**280,8**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Aug 1800**	
OA DB / WB	**31,1 / 21,3**	°C
Entering DB / WB	**24,1 / 17,1**	°C
Leaving DB / WB	**13,5 / 12,5**	°C
Coil ADP	**11,7**	°C
Bypass Factor	**0,150**	
Resulting RH	**52**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**206**	L/s
Standard L/s	**190**	L/s
Actual max L/(s-m²)	**19,78**	L/(s-m²)
Fan motor BHP	**0,14**	BHP
Fan motor kW	**0,11**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s
L/(s-m²)	**2,88**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-22

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-22**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **10,4** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	2,0	202	202	Aug 1800	0,9	10,4	19,39

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Sala de reunião 1	1	2,0	Aug 1800	202	0,9	10,4	19,39

Air System Design Load Summary for AC-22

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Aug 1800 COOLING OA DB / WB 31,1 °C / 21,3 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	31 m²	1231	-	31 m²	806	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	0 m²	0	-	0 m²	0	-
Ceiling	0 m²	0	-	0 m²	0	-
Overhead Lighting	101 W	62	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	374 W	353	-	0	0	-
People	3	195	195	0	0	0
Infiltration	-	0	0	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	184	19	10%	81	0
>> Total Zone Loads	**-**	**2024**	**214**	**-**	**887**	**0**
Zone Conditioning	-	1944	214	-	80	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	30	-	0	0	-
Return Fan Load	206 L/s	0	-	206 L/s	0	-
Ventilation Load	30 L/s	276	280	30 L/s	2	0
Supply Fan Load	206 L/s	114	-	206 L/s	-114	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	61	-	3%	27	-
>> Total System Loads	**-**	**2426**	**494**	**-**	**-5**	**0**
Central Cooling Coil	-	2426	495	-	0	0
>> Total Conditioning	**-**	**2426**	**495**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-22

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

August DESIGN COOLING DAY, 1800

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	31,1	0,01321	30	400	276	280
Vent - Return Mixing	Outlet	24,1	0,01028	206	867	-	-
Central Cooling Coil	Outlet	13,5	0,00939	206	867	2426	495
Supply Fan	Outlet	14,0	0,00939	206	867	114	-
Cold Supply Duct	Outlet	14,3	0,00939	202	867	-	-
Zone Air	-	22,9	0,00978	202	948	1944	214
Return Plenum	Outlet	23,1	0,00978	202	948	30	-
Duct Leakage Air	Outlet	14,0	0,00939	4	867	-	-
Return Duct	Outlet	22,9	0,00978	206	946	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	2024	Cooling	1944	22,9	202	948	0	0

Air System Sizing Summary for AC-23

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name	**AC-23**	
Equipment Class	**SPLT AHU**	
Air System Type	**SZCAV**	
Number of zones	**1**	
Floor Area	**24,5**	m²
Location	**Sao Paulo, Brazil**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	
Sizing Data	**Calculated**	
Zone L/s Sizing	**Sum of space airflow rates**	
Space L/s Sizing	**Individual peak space loads**	

Central Cooling Coil Sizing Data

Total coil load	**7,1**	kW
Sensible coil load	**5,7**	kW
Coil L/s at Jul 1500	**509**	L/s
Max block L/s	**509**	L/s
Sum of peak zone L/s	**509**	L/s
Sensible heat ratio	**0,800**	
m²/kW	**3,4**	
W/m²	**291,6**	
Water flow @ 5,6 °K rise	**N/A**	
Load occurs at	**Jul 1500**	
OA DB / WB	**33,2 / 21,9**	°C
Entering DB / WB	**23,4 / 16,9**	°C
Leaving DB / WB	**13,3 / 12,4**	°C
Coil ADP	**11,5**	°C
Bypass Factor	**0,150**	
Resulting RH	**55**	%
Design supply temp.	**13,0**	°C
Zone T-stat Check	**1 of 1**	OK
Max zone temperature deviation	**0,0**	°K

Supply Fan Sizing Data

Actual max L/s	**509**	L/s
Standard L/s	**471**	L/s
Actual max L/(s-m²)	**20,79**	L/(s-m²)
Fan motor BHP	**0,36**	BHP
Fan motor kW	**0,28**	kW
Fan static	**300**	Pa

Outdoor Ventilation Air Data

Design airflow L/s	**30**	L/s
L/(s-m²)	**1,22**	L/(s-m²)
L/s/person	**7,50**	L/s/person

Zone Sizing Summary for AC-23

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

Air System Information

Air System Name **AC-23**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **24,5** m²
Location **Sao Paulo, Brazil**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone L/s Sizing **Sum of space airflow rates**
Space L/s Sizing **Individual peak space loads**

Zone Sizing Data

Zone Name	Maximum Cooling Sensible (kW)	Design Airflow (L/s)	Minimum Airflow (L/s)	Time of Peak Load	Maximum Heating Load (kW)	Zone Floor Area (m²)	Zone L/(s-m²)
Zone 1	5,0	499	499	Jul 1500	0,8	24,5	20,37

Zone Terminal Sizing Data

No Zone Terminal Sizing Data required for this system.

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (kW)	Time of Load	Air Flow (L/s)	Heating Load (kW)	Floor Area (m²)	Space L/(s-m²)
Zone 1							
Copa 1	1	5,0	Jul 1500	499	0,8	24,5	20,37

Air System Design Load Summary for AC-23

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

	DESIGN COOLING			DESIGN HEATING		
	COOLING DATA AT Jul 1500 COOLING OA DB / WB 33,2 °C / 21,9 °C			HEATING DATA AT DES HTG HEATING OA DB / WB 8,9 °C / 4,4 °C		
ZONE LOADS	**Details**	**Sensible (W)**	**Latent (W)**	**Details**	**Sensible (W)**	**Latent (W)**
Window & Skylight Solar Loads	0 m²	0	-	0 m²	-	-
Wall Transmission	0 m²	0	-	0 m²	0	-
Roof Transmission	0 m²	0	-	0 m²	0	-
Window Transmission	0 m²	0	-	0 m²	0	-
Skylight Transmission	0 m²	0	-	0 m²	0	-
Door Loads	0 m²	0	-	0 m²	0	-
Floor Transmission	0 m²	0	-	0 m²	0	-
Partitions	24 m²	677	-	24 m²	363	-
Ceiling	25 m²	604	-	25 m²	323	-
Overhead Lighting	294 W	160	-	0	0	-
Task Lighting	0 W	0	-	0	0	-
Electric Equipment	1800 W	1621	-	0	0	-
People	4	207	240	0	0	0
Infiltration	-	1287	838	-	0	0
Miscellaneous	-	0	0	-	0	0
Safety Factor	10% / 10%	456	108	10%	69	0
>> Total Zone Loads	**-**	**5011**	**1186**	**-**	**755**	**0**
Zone Conditioning	-	4842	1186	-	1	0
Plenum Wall Load	0%	0	-	0	0	-
Plenum Roof Load	70%	0	-	0	0	-
Plenum Lighting Load	30%	88	-	0	0	-
Return Fan Load	509 L/s	0	-	509 L/s	0	-
Ventilation Load	30 L/s	349	245	30 L/s	262	0
Supply Fan Load	509 L/s	283	-	509 L/s	-283	-
Space Fan Coil Fans	-	0	-	-	0	-
Duct Heat Gain / Loss	3%	150	-	3%	23	-
>> Total System Loads	**-**	**5713**	**1431**	**-**	**3**	**0**
Central Cooling Coil	-	5713	1432	-	0	0
>> Total Conditioning	**-**	**5713**	**1432**	**-**	**0**	**0**
Key:	**Positive values are clg loads** **Negative values are htg loads**			**Positive values are htg loads** **Negative values are clg loads**		

System Psychrometrics for AC-23

Project Name: AGCO - TÉRREO 02/07/2024
Prepared by: Pedro AGA Silva 02:58

July DESIGN COOLING DAY, 1500

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (°C)	Specific Humidity (kg/kg)	Airflow (L/s)	CO2 Level (ppm)	Sensible Heat (W)	Latent Heat (W)
Ventilation Air	Inlet	33,2	0,01321	30	400	349	245
Vent - Return Mixing	Outlet	23,4	0,01038	509	594	-	-
Central Cooling Coil	Outlet	13,3	0,00935	509	594	5713	1432
Supply Fan	Outlet	13,8	0,00935	509	594	283	-
Cold Supply Duct	Outlet	14,1	0,00935	499	594	-	-
Zone Air	-	22,8	0,01022	499	606	4842	1186
Return Plenum	Outlet	22,9	0,01022	499	606	88	-
Duct Leakage Air	Outlet	13,8	0,00935	10	594	-	-
Return Duct	Outlet	22,8	0,01020	509	606	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1,207; At site altitude = 1,116 W/(L/s-K)
Air Density x Heat of Vaporization x Conversion Factor: At sea level = 2947,6; At site altitude = 2723,8 W/(L/s) Site Altitude = 661,0 m

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (W)	T-stat Mode	Zone Cond (W)	Zone Temp (°C)	Zone Airflow (L/s)	CO2 Level (ppm)	Terminal Heating Coil (W)	Zone Heating Unit (W)
Zone 1	5011	Cooling	4842	22,8	499	606	0	0